

概述 / SDN Overview

为什么需要SDN? / Why SDN?

- 网络管理困难 / Difficult network management
- 需要更简单的网络演进 / Need easier network evolution
- 打破厂商锁定 / Break vendor lock-in

SDN四个关键特征 / Four Key SDN Features

- 流平面控制转发 / Flow-controlled forwarding
 - 基于任意头部字段 (不仅是目的IP) / Based on any header fields (not just destination IP)
- 控制与数据平面分离 / Separation of control and data planes
- 网络控制功能外置 / Network control functions external
 - 控制器和应用在交换机外部 / Controller and apps external to switches
- 可编程网络 / Programmable network

OpenFlow协议 / OpenFlow Protocol

匹配/动作范式 / Match+Action Paradigm:

匹配 / Match: 头部字段 (MAC, IP, TCP/UDP端口号) / Header fields (MAC, IP, TCP/UDP ports, etc.)

动作 / Action: 转发 (forward), 丢弃 (drop), 修改 (modify), 发送到控制器 (send to controller)

链路层服务 / Link Layer Services

服务 Service	说明 Description
封装 Framing	将数据封装成帧, 添加头部、尾部 / Encapsulate datagram into frame, add header/tail
链路接入 Link Access	共享介质的信道接入, MAC地址识别源、目的 / Channel access for shared medium, MAC addresses identify source, destination
可靠传递 Reliable Delivery	通过确认和重传传递 / Via ACKs and retransmissions
错误检测 Error Detection	接收方检测错误 / Receiver detects errors
错误纠正 Error Correction	接收方识别并纠正错误 / Receiver identifies and corrects bit errors

6.4 Ethernet CSMA/CD 算法 / Ethernet CSMA/CD Algorithm

算法步骤 / Algorithm Steps

- NIC从网络接收数据, 创建帧 / NIC receives datagram from network layer, creates frame
- 如果NIC感知空闲信道, 开始传输: 如果忙, 等待直到空闲再传 / If NIC senses channel idle, start transmission; if busy, wait until idle
- 如果NIC在传输前检测到冲突, 中止 / If NIC detects collision before transmission, abort
- 如果NIC在传输过程中检测到冲突, 中止并发送退避信号 / If NIC detects other transmission, abort and send jam signal
- 中止后, NIC进入二进制指数退避 / Binary Exponential Backoff:
 - 第k次冲突后, 从0, 1, 2, ..., 2^k-1中随机选择一个 / After m-th collision, choose K randomly from {0, 1, 2, ..., 2^m-1}
 - NIC等待K*512比特时间, 返回步骤2 / NIC waits K*512 bit times, return to step 2
 - 更多冲突→更长退避时间 / More collisions → longer backoff interval
- 最大退避次数=16 / Maximum attempts = 16

三、Match + Action 抽象 (Match + Action Abstraction)

核心思想

对每个到达的数据包 (incoming packet):

- Match (匹配): 检查其包头字段 (header fields) 是否符合流表 (flow table) 中的某条规则 (rule)。
- 动作 / Action: 若匹配成功, 执行预定义动作 (predefined action), 如:
 - 转发到某端口 (forward to port)。
 - 丢弃 (drop)。
 - 修改 TTL (modify TTL) 等。

流表 (Flow Table) 包含:

- 匹配字段 (Match fields)
- 优先级 (Priority, 用于解决规则冲突 / resolve rule conflicts)
- 动作 (Action)
- 计数器 (Counters, 统计匹配的数据包数 / #packets 和字节数 / #bytes)

2. 示例 (见课件 Slide 4-130)

源IP (Source IP)	目的IP (Destination IP)	动作 (Action)
10.1.2.3	*	发送给控制器 (send to controller)

四、OpenFlow: Match + Action 的具体实现 (OpenFlow: Match+Action in Action)

1. OpenFlow 1.0 流表结构 (Slide 4-131)

匹配字段 (Match fields, 覆盖三层协议):

- 链路层 (Link layer):
 - 入口口 (ingress port)
 - 源/目的 MAC (src/dst MAC)
 - VLAN ID
 - 以太网型 (Ethernet Type)
- 网络层 (Network layer):
 - 源/目的 IP (src/dst IP)
 - IP协议 (IP protocol)
 - ToS (DSCP)
- 传输层 (Transport layer):
 - TCP/UDP 源/目的端口 (src/dst port)

动作类型 (Action types):

- 转发到指定端口 (forward to port(s)), 包括:
 - 洪泛 (flood)。
 - 发送给控制器 (send to controller)
- 丢弃数据包 (drop packet)
- 修改包头字段 (modify header fields), 如:
 - 改 MAC。
 - 改 IP。
 - 改封装 (rewrite port)。
- 封装并发送给控制器 (encapsulate and forward to controller), 用于处理未知流 (unknown)

2. 应用示例 (Slides 4-132 ~ 4-138)

- 目的IP转发 (Destination-based forwarding): 传统路由功能 (traditional routing)
- 防火墙 (Firewall): 阻止特定源IP或目的端口 (如 TCP 22 / SSH port)
- 二层交换 (Layer 2 switching): 基于 MAC 地址转发 (forward by MAC address)
- 负载均衡 (Load balancing): 根据源/目的IP选择不同路径 (path selection) → 传统IP路由无法实现

OpenFlow abstraction

Match + action abstraction unifies different kinds of devices

Router:

- match: longest destination IP prefix
- action: forward out a link

Switch:

- match: destination MAC address
- action: forward out a link

NAT:

- match: IP address and port
- action: rewrite address and port

图例: 展示了不同设备如何通过匹配和动作进行抽象。

OpenFlow protocol

operates between SDN controller and SDN-controlled switch

TCP used to exchange messages

- optional encryption
- three classes of OpenFlow messages:
 - controller-to-switch
 - asynchronous (switch to controller)
 - symmetric (misc)
- distinct from OpenFlow API
- AP used to specify generalized forwarding actions

OpenFlow protocol (续)

1. 什么是 802.11 标准?

- 物理层: 802.11 使用 ISM 频段 (Industrial, Scientific, Medical) 进行通信, 无需授权。
- 2.4 GHz 频段: 每个信道带宽 22 MHz, 全频段 2.4-2.4835 GHz
- 5 GHz 频段: 用于 802.11a/n/ac, 带宽更大, 干扰更少
- 6 GHz 频段: 802.11ax (Wi-Fi 6E) 新增, 提供更优 PPT 吞吐量

2. 信道划分: 信道分为 14 个信道 (各信道带宽不同, 中国/美国常用 1-11)

- 每个信道中心频率间隔 5 MHz
- 每个信道带宽为 20 MHz (802.11n/ac 可合并为 40/80/160 MHz)
- 信道: 信道资源有限

★ 关键点 (PPT 7-39):

- "A WiFi jungle is any physical location where a wireless station receives a sufficiently strong signal from two or more APs"
- 如果多个 AP 使用相同信道资源, 站点相互干扰, 导致吞吐量下降。

2.2.4 GHz 信道资源分配 (举例)

信道	中心频率 (MHz)	带宽 (MHz)
Ch 1	2412	2402-2422
Ch 6	2437	2427-2447
Ch 11	2462	2452-2472

只有 Ch 1, 6, 11 是互不重叠的信道 (间隔 ≥ 25 MHz)

若 AP 用 Ch 3, AP2 用 Ch 4 → 严重干扰

★ 管理责任 (PPT 7-40):

- "AP admin chooses frequency for AP. Interference possible: channel can be same as that chosen by neighboring AP"

2. 排队阻塞 (Queueing Delay, d_{queue})

- 重点: 在包到达队列中等待的时间。
- 特点:
 - 阻塞不确定, 阻塞的时间。
 - 网络拥塞 (阻塞), 延迟大, 延迟大, 延迟大。
 - 如果网络拥塞, 网络延迟会变大 (即“高延迟”)。
- 关键公式 (PPT 1-83):
$$L = \frac{\lambda}{\mu}$$
 - L: 队列长度 (bits)
 - λ: 平均到达率 (packets/sec)
 - μ: 平均服务率 (packets/sec)
 - 若 λ → 1, 排队延迟会变大; 若 λ > 1, 系统饱和, 排队延迟无穷大。

流量强度 $\frac{L}{\lambda}$

图例: 展示了排队阻塞的模型和公式。

CSMA/CA 协议流程 / CSMA/CA Protocol Process

发送方 / Sender:

- 如果感知信道空闲 (DIFS 时间) / If channel sensed idle for DIFS time:
 - 传输帧 (帧头 + 帧体) / Transmit entire frame (no CD)
- 如果感知信道忙 / If channel sensed busy:
 - 启动随机退避计数器 / Start random backoff timer
 - 信道空闲时计时器开始 / Timer counts down while channel idle
 - 信道忙时计时器暂停 / Timer freezes while channel busy
 - 计时器到期时传输 / Transmit when timer expires
 - 如果无 ACK, 增加帧重传次数, 重传次数 ≥ 2 / If no ACK, increase retransmission count

接收方 / Receiver:

- 如果帧接收正确 / If frame received OK:
 - 等待 SIFS 后返回 ACK / Return ACK after SIFS

帧间间隔 / Inter-Frame Spacing

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SIFS < DIFS: 确保 ACK/CTS 比数据帧有更高优先级 / Ensures ACK/CTS has higher priority than data frames

RTS/CTS 机制 / RTS/CTS Mechanism

解决隐藏终端问题 (Resolves Hidden Terminal Problem)

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注意 / Note:

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用户数据报 UDP 有两个字段: 数据字段和首部字段。

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图例: 展示了 UDP 报文的结构和字段。

CSMA/CA 协议流程 (续)

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注意

- 非持久连接 (Non-persistent HTTP)：默认模式 / Default mode
- 每个TCP连接最多传输一个对象 / At most one object per TCP connection
- 下载多个对象需要多个TCP连接 / Multiple TCP connections required to download multiple objects
- 响应时间 / Response time: 2RTT + 文件传输时间 / file transmission time (每个对象 / per object)

HTTP/1.1

- 持久连接 (Persistent HTTP)：默认模式 / Default mode
- 服务器端延迟后保持连接打开 / Server keeps connection open after sending response
- 同一客户端-服务器端的后续HTTP消息通过同一个连接发送 / Subsequent HTTP messages between same client/server sent over same connection
- 引入了流水线 (Pipelined GET)：在单个TCP连接上发送多个GET请求 / Send multiple GET requests on a single TCP connection
- 服务器按照FCFS (先来先服务 / First-Come-First-Served) 顺序响应GET请求 / Server responds to GET requests in FCFS order
- 问题 / Problems:
 - 队头阻塞 (Head-of-Line blocking, HOL blocking)：小对象需要等待大对象传输完成 / Small objects may have to wait for large object to finish transmission
 - 丢包恢复阻塞所有对象传输 / Packet loss recovery blocks all object transmissions

HTTP/2 [RFC 7540, 2015]

- 关键目标 / Key goal: 减少多对象HTTP请求的延迟 / Reduce latency for multi-object HTTP requests
- 主要改进 / Major improvements:
 - 方法、状态码、大多数头部与HTTP/1.1保持兼容 / Methods, status codes, most header fields unchanged from HTTP/1.1
 - 传输顺序基于客户端指定的对象优先级 (不一定是FCFS) / Transmission order based on client-specified object priority (not necessarily FCFS)
 - 服务器推送 (Server Push)：可以向客户端推送未请求的对象 / Can push unrequested objects to client
 - 帧分割 (Frame Division)：将对象分成帧，帧传输可以交错进行，缓解HOL阻塞 / Divide objects into frames, frame transmission can be interleaved, mitigating HOL
 - 二进制制式 (Binary format) (HTTP/1.x是ASCII制式 / HTTP/1.x is ASCII format)
 - 多路复用 (Multiplexed, 优先调度 / Prioritized)
- 问题 / Problem: 单个TCP连接上的传输阻塞问题仍然存在 / Transport-layer HOL blocking on single TCP connection still exists

HTTP/3:

- Transport layer changes from TCP to QUIC (UDP-based)
- Eliminates HOL blocking completely: each stream processed independently
- Faster connection setup: 1-RTT initial, 0-RTT resumption
- Connection migration: connections survive IP address changes
- Built-in TLS 1.3 encryption, more secure

三种类型的DNS服务器 / Three types of DNS servers:

- 1. 根DNS服务器 (Root DNS servers): 全球13个逻辑根服务器 / 13 logical root servers worldwide
- 2. 顶级域DNS服务器 (TLD servers / Top-Level Domain servers): 负责.com, org, net, edu等及国家顶级 / Responsible for
- 3. 权威DNS服务器 (Authoritative DNS servers): 组织自己的DNS服务器 / Organization's own DNS server(s)

本地DNS服务器 (Local DNS server)

- 不严格属于层次结构 / Does not strictly belong to hierarchy
- 每个ISP都有一个 / Each ISP has one
- 主机发起DNS查询时首先联系本地DNS服务器 / Host contacts local DNS server first when initiating DNS query
- 作为代理，将查询转发到成功层次 / Acts as proxy, forwards query into hierarchy
- A记录：域名到IPv4地址映射，如(example.com, 93.184.216.34, A)
- AAAA记录：域名到IPv6地址映射
- NS记录：指定哪个服务器负责该域名，如(com, dns1.com, NS)
- CNAME记录：别名，将一个名称指向另一个，如(www.example.com → example.com, CNAME)
- MX记录：邮件服务器，如(example.com, mail.example.com, MX)

Type	Name	Value
A	主机名 Hostname	IP地址 IP address
NS	域名 Domain name	该域名的权威DNS服务器主机名 Hostname of authoritative DNS server for this domain
CNAME	别名 Alias name	规范名称 Canonical name
MX	域名 Domain name	邮件服务器名称 Mail server name

Client/Server (客户端/服务器) 架构 / Client/Server Architecture

- 服务器 / Server:
 - 始终在线的主机 / Always-on host
 - 永久IP地址 / Permanent IP address
 - 通常在数据中心，便于扩展 / Often in data centers, for scaling
- 客户端 / Client:
 - 联系，与服务器通信 / Contact, communicate with server
 - 可能间歇性连接 / May be intermittently connected
 - 可能有动态IP地址 / May have dynamic IP addresses
 - 不直接相互通信 / Do not communicate directly with each other

典型应用 / Typical applications: HTTP, IMAP, FTP

文件分发时间 / File distribution time:

$$D_{tot} \geq \max \left\{ \frac{NF}{u_s}, \frac{F}{d_{min}}, \frac{F}{u_s + \sum u_i} \right\}$$

- 服务器客户端线性增长 / Grows linearly with N

Peer-to-Peer (P2P / 对等网络) 架构 / P2P Architecture

- 特点 / Features:
 - 没有始终在线的服务器 / No always-on server
 - 任意端直接通信 / Any end-to-end systems directly communicate
 - 对等体 (Peers) 间其他对等体提供服务，同时其他对等体提供服务 / Peers request service from other peers, provide service to other peers
 - 可扩展性 (Scalability)：对等体带来的服务能力服务需求 / New peers bring new service capacity as well as new service demand
 - 对等体间歇性连接，IP地址变化 / Peers intermittently connected, IP address change
 - 管理复杂 / Complex management
- 典型应用 / Typical applications: P2P 文件共享 / P2P file sharing
- 文件分发时间 / File distribution time:

$$D_{tot} \geq \max \left\{ \frac{F}{u_s}, \frac{F}{d_{min}}, \frac{NF}{u_s + \sum u_i} \right\}$$

- 网络吞吐量随节点数增长 (因为对等体贡献上带宽) / Grows logarithmically with N (because peers also contribute upload bandwidth)

CDN概述 / CDN Overview

- CDN2可在互联网各地安装缓存分布的服务器 / CDN companies deploy geographically distributed cache servers across
- 将内容本地化，减少流量 / Localize content, reduce traffic
- 共享CDN / Shared CDN: 如Akamai, Limelight
- 专用CDN / Dedicated CDN: 如Google, Netflix

CDN部署策略 / CDN Deployment Strategies

Enter Deep (深入部署)

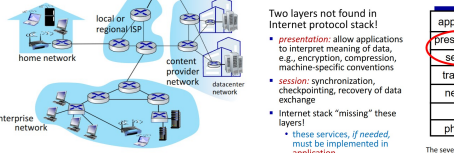
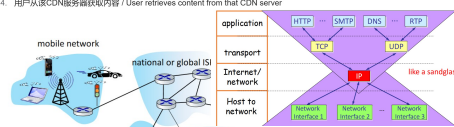
- 将CDN服务器部署到接入网络深处 / Deploy CDN servers deep into access networks
- 靠近用户 / Close to users
- 由Akamai采用 / Used by Akamai

Bring Home (集中部署)

- 将CDN服务器部署在少量关键位置 (如IXP / Internet Exchange Points) / Deploy CDN servers at fewer, key locations (like IXP)
- 较少服务器数量 / Fewer servers
- 由Limelight和其他CDN采用 / Used by Limelight and other CDNs

CDN内容访问流程 / CDN Content Access Process

- 1. 用户访问网页 / User accesses web page
- 2. DNS服务器重定向到CDN / DNS query is redirected to CDN
- 3. CDN返回最近最佳CDN服务器的IP / CDN returns IP of nearest/best CDN server
- 4. 用户从该CDN服务器获取内容 / User retrieves content from that CDN server



UDP工作流程:

1. 服务器: 创建socket, 绑定到端口, 等待数据
2. 客户端: 创建socket, 直接发送数据包 (含服务器地址和端口)
3. 服务器: 接收数据, 获取客户端地址, 回复
4. 客户端: 接收回复

```
python
1 # UDP客户端程序
2 from socket import *
3 servername = 'hostname'
4 serverport = 12000
5 clientsocket = socket(AF_INET, SOCK_DGRAM) # 创建UDP套接字
6 message = input('输入小写字母句子: ')
7 clientsocket.send(message.encode()) # 将消息封装成数据包
8 modifiedMessage, serverAddress = clientsocket.recvfrom(2048) # 接收服务器回复
9 print(modifiedMessage.decode())
10 clientsocket.close()
11
12 # UDP服务器程序
13 from socket import *
14 serverport = 12000
15 serversocket = socket(AF_INET, SOCK_DGRAM) # 创建UDP套接字
16 serverAddress, message = serversocket.recvfrom(2048) # 接收客户端
17 print('服务器接收到了: ', message)
18 while True:
19     message, clientAddress = serversocket.recvfrom(2048) # 接收数据
20     modifiedMessage = message.decode().upper()
21     clientsocket.send(modifiedMessage.encode(), clientAddress) # 回复
```

BitTorrent像一个文件交换俱乐部: / BitTorrent operates like a file exchange club:

基本组件: / Basic Components:

- 文件分片: 大文件分成256KB小块 / File chunking: Large files split into 256KB chunks
- Tracker: 记录谁有什么块的中央注册表 / Tracker: Central registry recording which peers have which chunks
- Torrent: 参与分享同一文件的用户群 / Torrent: Group of peers sharing the same file

工作流程: / Workflow:

- 你加入一个torrent, 从tracker获取其他用户列表 / You join a torrent, getting peer list from tracker
- 你连接到几个“邻居”用户 / You connect to several “neighbor” peers
- 你请求你没有的块, 优先请求稀有块 (防止某些块消失) / You request missing chunks, prioritizing rarest chunks first (to prevent scarcity)
- 你上传块给别人, 采用“以牙还牙”策略: / You upload chunks using “tit-for-tat” strategy:
 - 优先上传给你最多数据的4个用户 / Prioritize top 4 peers uploading most data to you
 - 每10秒重新评估最佳伙伴 / Re-evaluate best partners every 10 seconds
 - 每30秒随机服务一个新用户 (“乐观非阻塞”) , 给新用户机会 / Randomly serve a new peer every 30 seconds (“optimistic unchoking”) to give newcomers opportunities

Round Robin (轮询调度)

- arriving traffic classified, queued by class
- any header fields can be used for classification
- server cyclically, repeatedly scans class queues, sending one complete packet from each class (if available) in turn
- if one class to be served is found none, immediately check the next class
- unfair to short packets and has no priority
- 丢包策略 (当缓冲区满时):
 - 尾丢弃 (Tail Drop): 新的包直接丢 - 简单但导致TCP同步 (多个流同时降速)
 - Tail Drop: Drop arriving packets - simple but causes TCP synchronization (many flows slow down simultaneously);
 - 主动队列管理 (AQM) 如RED: 提前随机丢包, 通知发送方减速, 避免拥塞

Attribute	IBGP	EBGP
Scope	Within a single AS	Between different ASes
Route advertisement	Limited to IBGP peers	Broadcast to all peers
Topology	Requires full mesh (or route reflector)	No full mesh needed
Administrative Distance	200	20
Loop Prevention	IBGP split horizon	AS Path

- In IBGP all routers traditionally form a full mesh, directly peering with each other. Alternatively, a route reflector configuration can be used where some servers serve as route reflectors to reduce the need for full-mesh connections.

TCP工作流程:

1. 服务器: 创建“双socket”, 绑定端口, 监听连接
2. 客户端: 创建socket, 连接到服务器
3. 服务器: 接受连接, 为客户端创建新的“连接socket”
4. 双方通过连接socket交换数据
5. 通信结束后关闭连接socket (双socket保持打开, 等待新连接)

```
python
1 # TCP服务器程序
2 from socket import *
3 servername = 'servername'
4 serverport = 12000
5 clientsocket = socket(AF_INET, SOCK_STREAM) # 创建TCP套接字
6 clientsocket.connect((servername, serverport)) # 连接到服务器
7 sentence = input('输入小写字母句子: ')
8 clientsocket.send(sentence.encode()) # 通过连接socket发送数据
9 modifiedSentence = clientsocket.recv(1024) # 接收数据
10 print('收到回复: ', modifiedSentence.decode())
11 clientsocket.close()
12
13 # TCP客户端程序
14 from socket import *
15 serverport = 12000
16 serversocket = socket(AF_INET, SOCK_STREAM) # 创建TCP套接字
17 serversocket.bind(('', serverport)) # 绑定到端口
18 serversocket.listen(5) # 监听连接
19 print('服务器准备就绪: ')
20 while True:
21     connectionSocket, clientAddress = serversocket.accept() # 接受连接
22     sentence = connectionSocket.recv(1024).decode() # 从连接socket接收
23     modifiedSentence = sentence.upper()
24     connectionSocket.send(modifiedSentence.encode())
```

2.2 路由算法分类 / Routing Algorithm Classification

按信息来源分类 / By Information Source

- 集中式全局 (Link State) / Centralized/Global
 - 所有路由都有完整的路由和链路成本信息 / All routers have complete topology and link cost info
 - 链路状态算法 / Link State algorithm
- 分布式分散 (Distance Vector) / Decentralized/Distributed
 - 路由器只知道直接连接的邻居和到邻居的链路代价 / Routers know only physically connected neighbors and link costs
 - 迭代计算过程, 与邻居交换信息 / Iterative computation, exchange info with neighbors
 - 距离向量算法 / Distance Vector algorithm

3.3 Link-State算法 (Dijkstra算法) / Link-State Algorithm (Dijkstra's Algorithm)

特点 / Features:

- 集中式 / Centralized: 网络拓扑和链路代价对所有节点已知 / Network topology and link costs known to all nodes
- 通过链路状态广播 / Link State Broadcast实现
- 所有节点有相同信息 / All nodes have same info
- 计算从任一节点到所有其他节点的最短路径 / Computes shortest paths from one node to all other nodes

5.4 Distance Vector算法 (Bellman-Ford算法) / Distance Vector Algorithm (Bellman-Ford)

基础: Bellman-Ford方程 / Foundation: Bellman-Ford Equation

$$D_x(y) = \min_c [c_{x,c} + D_c(y)]$$

- 从x到y的最小代价 = 所有邻居中, x到邻居的代价加上邻居到y的最小值 / Minimum cost from x to y = minimum over all neighbors of cost from x to neighbor + minimum cost from neighbor to y

算法特点 / Algorithm Features

- 分布式 / Distributed: 每个节点只与直接邻居通信 / Each node communicates only with direct neighbors
- 迭代 / Iterative: 节点在本地接收代价变化或收到邻居DV更新的数据 / Nodes iterate when local link cost changes or neighbor DV update received
- 异步 / Asynchronous: 节点不需要同步 / Nodes don't need to synchronize
- 自终止 / Self-terminating: 没有问题需要更新时动作 / No action needed when no updates

“坏消息传得很慢” (无穷计数问题) / “Bad News Travels Slowly” (Count to Infinity Problem)

当链路代价增加时, 可能导致计数数据无穷问题 / When link cost increases, may result in count to infinity problem

毒性反转 / Poisoned Reverse

- 如果一条链路代价无穷 / If a route to x through y
- 告诉它到x的距离是无穷大 / z tells y its distance to x is infinity
- 部分解决问题 / Only partially solves problem

5.7 ICMP (互联网控制消息协议 / Internet Control Message Protocol)

概述 / Overview

- 用于主机和路由器之间通信网络层信息 / Used for communication of network-level info between hosts and routers
- 错误报告 / Error reporting: 不可达主机网络/端口协议 / Unreachable host/network/port/protocol
- 回声请求/应答 / Echo request/reply: ping/using by ping

常见ICMP类型 / Common ICMP Types

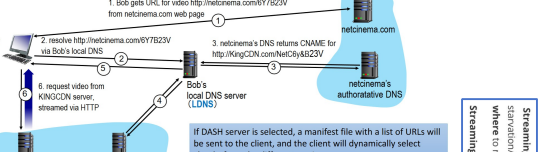
类型 Type	代码 Code	描述 Description
0	0	回声应答 Echo reply (ping)
3	0	目标网络不可达 Destination network unreachable
3	1	目标主机不可达 Destination host unreachable
3	3	目标端口不可达 Destination port unreachable
8	0	回声请求 Echo request (ping)
11	0	TTL超时 TTL expired

Traceroute与ICMP / Traceroute and ICMP

- 发送方发送TTL=1,2,3...的UDP数据包到目的地 / Sender sends UDP segments with TTL=1,2,3... to destination
- 每个路由器通过TTL超时返回ICMP TTL过期消息 / Each router returns ICMP TTL expired when TTL reaches 0
- 利用用的时间, 端口不可达返回ICMP消息 / When receiving destination, port unreachable returns ICMP message

Bob (client) requests video http://netcinema.com/67B23V

- video stored in CDN at http://KingCDN.com/NetCty&B23V



- CLI (Command-Line Interface)
 - Approach: Administrators log into individual devices directly via SSH or Telnet and enter configuration commands manually.
 - Drawbacks: Highly error-prone; does not scale; commands are vendor- and device-specific, making automation difficult.
 - Use Case: Suitable for small networks or emergency troubleshooting.
- SNMP/MIB (Simple Network Management Protocol / Management Information Base)
 - Architecture: Consists of a manager (management station) and multiple managed devices (naming agents).
 - MIB: A hierarchical, standardized database that defines all manageable data objects on a device (e.g., interface status, CPU utilization, routing table entries). Each object is uniquely identified by an OID (Object Identifier).
 - SNMP Protocol: The manager uses messages like GetRequest and SetRequest to query or modify values in the MIB. Devices can proactively send Trap messages to notify the manager of significant events.
 - Role: Primarily used for monitoring and status collection; limited in large-scale configuration management.
- NETCONF/YANG
 - Objective: Addresses the limitations of SNMP in configuration management by providing a more robust, secure, and transaction-oriented net. management framework.
 - NETCONF: An RPC (Remote Procedure Call)-based protocol that encodes messages in XML. It operates over secure transport layers (e.g., SSH, TLS) and supports atomic configuration operations such as create, update, and delete.
 - YANG: A data modeling language used to define the structure, syntax, and constraints of configuration and operational data managed by NETCONF. YANG models enable precise and validated configuration operations, enhance security through data validation, and support multi-vendor interoperability.

五大 RIR 分别是哪些?

- ARIN — American Registry for Internet Numbers
- RIPE NCC — Réseaux IP Européens Network Coordination Centre
- APNIC — Asia-Pacific Network Information Centre
- LACNIC — Latin America and Caribbean Network Information Centre
- AFRINIC — African Network Information Centre

ICANN和IANA关系 / ICANN sets up the overall system and delegates operational responsibilities to IANA.

IANA 的主要职责是什么? / IANA operates at the highest international level and coordinates global IP-related responsibilities. It further delegates work to five Regional Internet Registries (RIRs).

	类型	原理	通信速率	现代使用情况
内存交换	Memory-based switching	CPU把从输入内存复制到输出内存	内存带宽 (一次只复制一个包)	✗ 已淘汰 (太慢)
共享总线交换	Bus-based switching	所有包通过一条共享总线传输	总线带宽 (一次只复制一个包)	✗ 仅用于低端设备
交叉网络交换 (如Crossbar)	Interconnection-network switching (e.g., Crossbar)	多条并行总线, 类似快速交换器	Bus bandwidth (one packet at a time)	✗ 几乎无应用 (可并行)
多平面交换	Multiple parallel planes, like a subway interchange	Nearly no bottleneck (parallelism)		✗ Dominant in modern high-speed